2 Chapter 2: Time-Domain Analysis of Continuous Time Systems

The majority of linear systems that we will deal with here may be defined by a linear differential equation of the form

$$\frac{d^N y(t)}{dt^N} + a_1 \frac{d^{N-1} y(t)}{dt^{N-1}} + \ \dots \ + a_{N-1}\frac{dy(t)}{dt} + a_N y(t) =$$
$$b_{N-M}\frac{d^M x(t)}{dt^M} + b_{N-M+1}\frac{d^{M-1} x(t)}{dt^{M-1}} + \ .. \ + b_{N-1}\frac{dx(t)}{dt} + b_N x(t).$$

In other words, the input and output of a linear system are often described by N-th and M-th order linear differential equations, with $M \leq N$ in most cases. A *linear* differential equation is simply one in which each derivative is multiplied by a unique coefficient a_N or b_N.

The above can be simplified by using $\frac{d^N}{dt^N} = D^N$, which then reduces to

$$(D^N + a_1 D^{N-1} + \ \dots \ + a_{N-1}D + a_N)y(t) =$$
$$(b_{N-M}D^M + b_{N-M+1}D^{M-1} + \ \dots\ b_{N-1}D + b_N)x(t).$$

2.1 Zero-Input Response

Based on the previous discussion (Chapter 1) on the superposition principle, we can separate the output of a linear differential equation into two separate components, the zero-input response, and the zero-state response. Obeying the superposition principle, this means the total output may be found by summing the two corresponding inputs.

The first of these that we will solve for is the zero-input response, that is, the output $y_1(t)$ when $x_1(t) = 0$. In most systems, this is an important component in the system's input, corresponding to the initial state of the system at $t = 0$ *prior* to any real-time or future input.

By reducing the input to 0, then the associated linear differential equation reduces to

$$(D^N + a_1 D^{N-1} + \ \dots \ + a_{N-1}D + a_N)y_0(t) = Q(D)y_0(t) = 0,$$

with $y_0(t)$ representing the zero-input response.

Solving for $y(t)$, for all possible values of t, the left-hand side of the equation can only resolve to zero if $y_0(t)$ and all its derivatives are of the same form. The exponential function, $e^{\lambda t}$ meets this criteria, since $\frac{d}{dt}e^{\lambda t} = ae^{\lambda t}$. Setting each resulting coefficient a to Using $y_0(t) = e^{\lambda t}$, the equation now reduces to

$$c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} + c_N e^{\lambda_N t} = 0.$$

Now solving for $Q(\lambda) = 0$ (we'll save the constants c_N for the next step),

$$Q(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2)\ \dots\ (\lambda - \lambda_N) = 0,$$

where the values of λ are the characteristic roots, eigenvalues, or natural frequencies of the system.

Finally, the N coefficients may be solved for using N initial conditions, $y_1(0)$, $\frac{d}{dt}y_1(0)$, $\dots$ $\frac{d^N}{dt^N}y_1(0)$.